Section 3.2 Product and Quotient Rules

$$[A \cdot B]' = A' \cdot B + A \cdot B'$$

Ex ① $f(x) = \underbrace{x^2}_{A} \cdot \underbrace{(x+4)^3}_{B}$

$$A' \cdot B \quad + \quad A \cdot B'$$

$$f'(x) = 2x \cdot (x+4)^3 + x^2 \cdot 3(x+4)^2$$

$$f'(x) = x(x+4)^2 [2(x+4) + 3x]$$

$$f'(x) = x(x+4)^2 (5x+8)$$

Ex ② $y = \sqrt{x} \cdot e^x$

$$y' = \frac{1}{2\sqrt{x}} \cdot e^x + \sqrt{x} \cdot e^x$$

$$y' = e^x \left(\frac{1}{2\sqrt{x}} + \sqrt{x}\right)$$

Ex ③ Where does $y' = 0$ for

$$y = (x-2)^4(x+1)^5$$

$$y' = 4(x-2)^3 \cdot (x+1)^5 + (x-2)^4 \cdot 5(x+1)^4$$

$$0 = (x-2)^3(x+1)^4\left(4(x+1) + 5(x-2)\right)$$

$$0 = (x-2)^3(x+1)^4(9x-6)$$

$(x-2)^3 = 0$ | $(x+1)^4 = 0$ | $(9x-6) = 0$

$x-2=0$ | $x+1=0$ | $9x = 6$

$x = 2$ | $x = -1$ | $x = \frac{6}{9}$

$x = \frac{2}{3}$

Quotient Rule

$$\left[\frac{A}{B}\right]' = \frac{A' \cdot B - A \cdot B'}{B^2}$$

Ex 4 $f(x) = \dfrac{e^x}{x^2}$

$$f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{(x^2)^2}$$

$$f'(x) = \frac{\cancel{x} \cdot e^x(x-2)}{x^{\cancel{4}3}}$$

$$\boxed{f'(x) = \frac{e^x(x-2)}{x^3}}$$

Ex 5 $y = \dfrac{2x+5}{3x^2+1}$

$$y' = \frac{2 \cdot (3x^2+1) - (2x+5)(6x)}{(3x^2+1)^2}$$

$$y' = \frac{6x^2+2-12x^2-30x}{(3x^2+1)^2}$$

$$y' = \frac{-6x^2 - 30x + 2}{(3x^2+1)^2}$$

Ex ⑥ $g(x) = \dfrac{x+8}{\sqrt{x}}$

$$g'(x) = \frac{\left[1 \cdot \sqrt{x} - (x+8)\cdot\left(\frac{1}{2\sqrt{x}}\right)\right] 2\sqrt{x}}{\left[\sqrt{x}^{\,2}\right] 2\sqrt{x}}$$

$$g'(x) = \frac{2x - x - 8}{x \cdot 2\sqrt{x}}$$

$$g'(x) = \frac{x-8}{2x\sqrt{x}}$$